AKAASH BELLA

Portland, Oregon | | akaash.bella@gmail.com | LinkedIn

EDUCATION

Oregon State University | Corvallis, OR

- M.S. in Computer Science — Expected Dec 2027
- B.S. in Computer Science & B.S. in Innovation Management (Double Degree); Minor in Business — Sep 2021 – Dec 2025

TECHNICAL SKILLS

- **Languages:** Python, C++, C, SQL, JavaScript
- **ML & Data Science:** PyTorch, Torchvision, TensorFlow, scikit-learn, NumPy, pandas, Matplotlib, CUDA
- **Tools & DevOps:** Git, GitHub, Jupyter, Linux, MongoDB, Slurm, LaTeX, Cursor, Codex, Agile/Scrum

RESEARCH EXPERIENCE

Graduate Researcher | Oregon State University — Jan 2026 – Present

Weight Perturbation Sensitivity in Convolutional Neural Network Training

- Investigate how weight perturbation magnitude during training affects CNN classification accuracy and generalization across architectures.
- Use Torchvision for CNN architectures and dataset pipelines; implement controlled Gaussian weight perturbations at multiple noise strengths during training.
- Use NumPy, pandas, and Matplotlib to process experimental metrics and visualize classification accuracy and loss across perturbation strengths.

ACADEMIC & PROJECT EXPERIENCE

Graduate Teaching Assistant | Oregon State University — Jan 2026 – Present

- Mentored 400+ students in Open-Source Software (CS464) and Usability Engineering (CS352), providing technical guidance on software architecture, UI/UX principles, and HCI implementation.
- Evaluated technical reports and code contributions, ensuring students adhered to industry-standard documentation and professional coding practices.
- Served as GTA and course designer for BA 363 (Innovation Management; 100+ students), handling TA duties and revamping lectures, materials, homework, and exams for the following year.

ResearchSeek | Independent Technical Project — Oct 2026

Linux Semantic Search Engine

- Built a Linux command-line search tool using Python transformer embeddings, a C++ retrieval engine, and SQLite storage.
- Implemented cosine-similarity search with heap-based top-k retrieval, incremental indexing, and benchmarks comparing semantic and keyword search.

Capstone Project Lead | AgBizLogic (AI/UX Project) — 2024–2025 | Link

- Led a cross-functional team over three terms to develop and manage the full lifecycle of an AI Wizard-of-Oz prototype for grower decision-support tools.
- Validated system usability and user trust through iterative testing, delivering a strategic product roadmap and UX recommendations to AgBizLogic stakeholders.

Consulting Project | Mauka Energy (FEVER/BEVER) — 2025 | Link

- Architected a commercialization strategy for regenerative braking and routing tools by integrating technical performance data with market demand analysis.
- Developed detailed financial models (NPV, Payback Period) to justify investment viability and presented strategic recommendations to executive leadership.

RELEVANT COURSEWORK

- **Computer Science & AI:** Machine Learning; AI Agents, Search, and Reasoning; AI Optimization; Algorithms; Data Structures; Database Management Systems; Software Engineering Methods; Operating Systems.
- **Innovation & Management:** Innovation Management; Innovation and New Product Development; Technology Commercialization; Managing Ethics and Corporate Social Responsibility; Managing Organizations; Leadership.

AWARDS

- Dean's List (6-time recipient) | Oregon State University
- State Retail Champion | Oregon DECA State Career Development Conference, Sunset High School